# Approval-Based Elections in the 1D Euclidean Domain

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# TODO

## 0.1 Resources

https://github.com/jasieksz/master-thesis-code

## 0.2 My Questions

1. Plan pracy
2. co z Chińczykami, dówód / obalić
3. VCR MAV - DP, zacząć od rekurencji, branch and bound, struktura danych?
4. Jak pisać - czas / osoba

## 0.3 10/12/2020

1. Refactor code - vcr ilp detection, profile generation
2. Generate and analyze profiles: 5x5, 4x6, 6x4, random NxM

## 0.4 17/12/2020

1. Implement ilp MAV
2. Implement VR/CR MAV **?**
3. Compare ilp and poly MAV

## 0.5 07/01/2021

1. VR/CR MAV **?**

## 0.6 14/01/2021

1. VR/CR MAV **?**

## 0.7 11/02/2021

1. write thesis (+- 3 pages)
2. email about CR MAV to Liu/Guo
3. fix CR MAV algo
4. try to proof CR MAV
5. single thread experiments

## 0.8 18/02/2021

1. investigate CR MAV
2. VCR MAV ideas

3. email about CR MAV to Liu/Guo
4. try to proof CR MAV
5. write thesis
6. 20x20 VCR profiles - random distribution analysis
7. delete some random Candidates for VCR profile to become VR / CR

## 0.9 25/02/2021

1. investigate CR MAV
2. VCR MAV ideas
3. write thesis

## 0.10 04/03/2021

1. Chińczycy - wyslac kontrprzykład
2. VCR MAV ideas
3. write thesis - ch.3 (1D Euclidean Pref.) and ch.5 (domain analysis)
4. 20x20 VCR profiles - random distribution analysis
5. delete some random Candidates for VCR profile to become VR / CR
6. Control problem for VCR
7. Justified Representation (Proportional / Extended) for VCR

## 0.11 11/03/2021

1. VCR MAV ideas
2. write thesis - ch.6 MAV

# Abstract

(WIP)

I study approval-based elections, where preferences are in the 1D-Euclidean domain. I explore multiple areas, notably:

1. I create domain detection algorithms and evaluate the results.
2. I look for the efficient method of computing election results.

For the first issue I find ???. For the second one I create polynomial complexity algorithm (not yet) and compare it with LP solutions. (WIP)

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# Chapter 1

# Introduction

Elections are ubiquitous element of our society in one form or another. For example choosing members of parliament or a winner of a competition. In a day-to-day activities we need to decide on a meeting time or select recommended movies in a streaming service. Each of these choices may require different type of election and scoring rule, based on the underlying properties of the choice (i.e. voters should be represented proportionally by members of a government)

### Reseach objectives

In this work I am going to focus on the *Approval Elections*, that is where each voter provides a subset of candidates that he approves of and disapproves the rest. In general many social choice problems are known to be computationally hard. However they become tractable under some restrictions on the structure of the preferences. Elkind and Lackner (2015) explored well known restrictions in the setting of approval preferences, introduced several new one-dimensional domain restrictions such as Dichotomous Euclidean (sprawdzic kiedy/kto wprowadził pojęcie - spatial pref, multidim unfolding) and derivatives (czy to dobre okreslenie na VR CR?). Faliszewski (2020) generalized this model for the N-dimensional euclidean space calling it Voter Candidate Range and explored two-dimensional case.

The goal of this work is to explore properties of the one-dimensional model. Search for the profile detection polynomial complexity algorithm. I will try as well to extend solution of **?** for computing MAV score with polynomial complexity to the VCR model.

### Contributions

I provide linear programming preference detection algorithm and ...

# Chapter 2

# Preliminaries

In this chapter, we introduce the definitions and notations used in this work.

### Approval Elections

Election is a pair $E = (C, V)$, where $C = \{c_1, ..., c_m\}$ is a set of candidates and $V = \{v_1, ..., v_n\}$ is a set of voters. Each voter $v \in V$ approves a subset of the candidates and disapproves the remaining candidates. I will be referring to it as a *approval set* and for each voter $v \in V$ denote it as $A(v)$, when using $A$ just by itself I refer to all approval sets in the given election (preferences).

### Committee Selection Rules

Voting rules are algorithms used to select winners (committee) of a given election. Each rule has different properties and must be selected accordingly to the goal of the election. For the detailed overview of multiwinner voting rules I point to Faliszewski et al. (2017).

### WIP: Jakie jeszcze tematy/pojęcia tu opisać

# Chapter 3

# One Dimmensional Euclidean Domain

## 3.1 Preference Restrictions

(WIP: Motywacja, po co restrykcje, czy moge powtorzyc Introduction?)

Many social choice problems such as winner determination, (WIP: Control?) are known to be computationally hard. To deal with it we can use approximate or parametrized algorithms (WIP: Podać przykłady metod / cytacje). The alternative approach is to introduce restrictions on the preference domains. (WIP: Cytacje research'u) In the last few years this area of research advanced a lot, providing multiple efficient algorithms for the popular committee selection rules, and other problems.

## 3.2 Euclidean Preference Restrictions

### Voter Candidate Range

Given an election $E = (C, V)$, we say that the voters have *1D-Euclidean* preferences if for every agent (voter or candidate) there exists a point $x_a$ and nonnegative real value radius $r_a$ such that: $\forall\, c \in C\ \forall\, v \in V : c \in A(v) \Leftrightarrow \sqrt{(x_c - x_v)^2} \leq r_c + r_v$

We can easily think of it as: the point $x_a$ represents ideal position in the 1-dimensional space of opinions. The radius can be seen as candidate's charisma or voter's willingness to compromise.

### Candidate Range

We say that P satisfies CR if candidates in $C$ can be ordered so that each of the sets $v_i$ forms an interval of that ordering. CR is special case of VCR, where we require all the voters to have radii equal to zero

### Voter Range

We say that P satisfies VR if the voters in $V$ can be reordered so that for every candidate $c_i$ the voters that approve c form an interval of that ordering. VR is a special case of VCR, where we require all the candidates to have radii equal to zero

### Voter Interval and Candidate Interval

Faliszewski (2020) showed that CR and VR are equivalent to respectively VI and CI introduced by Elkind and Lackner (2015).

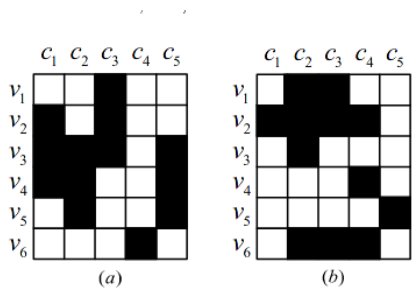


Fig. 3.1 CR and VR Temporary image

### Larger Domain

(WIP: Zmienić na mój przykład i opisać dlaczego nie vr i nie cr)

The VCR model has larger domain than the VR and CR combined. For example, election with candidate set $C = \{a,b,c,d\}$ and voters with approval sets $\{a,b\}$, $\{b,c\}$, $\{b,d\}$, and $\{a,b,c,d\}$ is VCR, but neither VR nor CR.

# Chapter 4

# Voter Candidate Range Profile Detection

This chapter describes in detail and compares created algorithms for the VCR property detection.

## 4.1 Profile Structure Detection

Detection algorithms are used to check whether election profile satisfies desirable structural properties. For example to check CR (VR) properties it's enough to check for the *consecutive ones property* in the preferences matrix.

Our focus lies on the one dimmensional VCR property. Having preferences matrix as input we have to provide a mapping of each agent (voter/candidate) to the point on the real line and nonnegative number representing radius, so that it satisfies the VCR property.

## 4.2 Integer Linear Programming

ILP is a common approach to solve NP-hard problems, furthermore it still can be used effictively for other problems. First we have to define the problem as an integer linear program. That is a system of linear inequalities and constraints and optionally linear function to optimize, alternatively we can just check feasibility of the constraints. To obtain a solution we have to use an optimization solver. Popular choices are Gurobi or CPLEX which are state of the art general purpose solvers.

Solution based on the ILP is excellent choice for the first algorithm to develop. It provides efficient and relatively simple detection algorithm. Additionally it can be used as a fast benchmark for checking future solutions.

### Transform VCR definition into linear constraints

Before forumulating ILP we have to redefine the VCR definition beacasue it is based on the euclidean distance hence contains quadratic function. Equation 4.1 shows transforamtion of VCR equation into linear inequalities.

$$\begin{aligned} VCR &\iff \sqrt{(x_c - x_v)^2} \leq r_c + r_v \\ &\iff \|x_c - x_v\| \leq r_c + r_v \\ &\iff -(r_c + r_v) \leq x_c - x_v \leq r_c + r_v \\ &\iff x_c - x_v \leq r_c + r_v \ \ \vee \ \ -(x_c - x_v) \leq r_c + r_v \end{aligned} \tag{4.1}$$

The constraint obtained from equation 4.1 is not suitable for ILP yet beacasue it contains logical *OR* condition which cannot be directly represented in the ILP. However this issue is quite common and most solvers provide additional mechanisms to solve it.

In Gurobi solver it's called *INDICATOR* constraint which allows to create versatile constraints like: $z = f \rightarrow a^T x \leq b$ which states that if the binary indicator variable $z$ is equal to $f$, where $f \in \{0,1\}$, then the linear constraint $a^T x \leq b$ should hold. On the other hand, if $z = 1 - f$, the linear constraint may be violated.

### Detection algorithm

Final formulation of the ILP for recognising VCR property in the election profile is shown in figure 4.2. Variables used by the program $x_a$ and $r_a$ represent agent's point on the real line and their radius respectively. Additionally we use two binary variables $p$ and $q$ required by the indicator constraints.

(WIP: Czy pisać jeszcze raz że approval set $A$ jest jedyną daną wejściową?)

(WIP: Wyjaśnić rodzaje zmiennych w ILP? (binary, etc.))

(WIP: Poprawić format zapisu ILP na bardziej standardowy)

**check feasibility** :

**subject to** :

$$
\begin{aligned}
& x_a \in \mathbb{R} && \forall \text{ a} \in C \cup V \\
& r_a \in \mathbb{R}^{0+} && \forall \text{ a} \in C \cup V \\
& p_{cv} \in \{0,1\} && \forall \text{ c} \in \text{C} \; \forall \text{ v} \in V \\
& q_{cv} \in \{0,1\} && \forall \text{ c} \in \text{C} \; \forall \text{ v} \in V \\
\text{if } c \in A(v) : & && \\
& x_c - x_v \leq r_c + r_v && \\
& -(x_c - x_v) \leq r_c + r_v && \\
\text{if } c \notin A(v) : & && \\
& p_{cv} == 1 \rightarrow x_c - x_v \geq r_c + r_v + 1 && \\
& q_{cv} == 1 \rightarrow -(x_c - x_v) \geq r_c + r_v + 1 && \\
& p_{cv} + q_{cv} >= 1 &&
\end{aligned}
\tag{4.2}
$$

## 4.3 (WIP) My VCR detection algorithm

# Chapter 5

# Voter Candidate Range Profiles

In this section we explore VCR profiles domain and describe found statistical properties. The main goal is to compare size of the domain with other well known domains such as VR and CR.

## 5.1 Domain analysis

We've performed several experiments. (WIP: tense - present perfect vs past simple vs present?) For the first one we check how many profiles have VCR property but not VR nor CR. As the input we use every combination (WIP: czy to na pewno kombinacje?) of the profiles for the given election size. For the second experiment we generate random VCR profiles and again check how many of them are not VR nor CR. In both experiments we use detection algorithm described in the section 4.2.

All the resources and implemention details related to the task are included in the appendix **??**.

## 5.2 (WIP) How many VCR profiles are not VR nor CR?

**Experiment 1 - small profiles**

For the small profiles we performed a complete search of the domain. We checked the elections with the following sizes ($M$ candidates $N$ voters): $[(3,3),(4,4),(5,5),(4,6),(6,4)]$. The results as seen in the table 5.1 show that VCR domain is relatively large, however only limited number of profiles are not VR nor CR.

WIP : More advanced conclusions?

WIP : Prettify the table

| | 3C 3V | 4C 4V | 5C 5V | 4C 6V | 6C 4V |
|---|---|---|---|---|---|
| Count | $2^9$ | $2^{16}$ | $2^{25}$ | $2^{24}$ | $2^{24}$ |
| VCR | 506 | 57832 | 19846082 | 10893736 | 10893736 |
| VR | - | - | - | - | - |
| CR | - | - | - | - | - |
| VCR !COP | 0 | 96 | 501000 | 144480 | 144480 |

Table 5.1 Temporary table - Small profiles NCOP/VCR

WIP : Make some charts, write statistics, show examples

### Profile examples

Example of 5C 5V election that is VCR but not VR nor CR.

WIP: Replace with a drawing and explain why not VR/CR

```
Approval Set
    [1., 1., 0., 0., 1.]
    [1., 1., 1., 1., 1.]
    [1., 0., 0., 0., 0.]
    [0., 0., 0., 1., 1.]
    [0., 0., 1., 0., 1.]

Candidates:
    Candidate(id='A', x=3.0, r=1.0),
    Candidate(id='B', x=2.0, r=0.0),
    Candidate(id='C', x=0.0, r=0.0),
    Candidate(id='D', x=1.0, r=0.0),
    Candidate(id='E', x=1.0, r=1.0)

Voters:
    Voter(id='V0', x=2.0, r=0.0),
    Voter(id='V1', x=0.0, r=3.0),
    Voter(id='V2', x=3.0, r=0.0),
    Voter(id='V3', x=1.0, r=0.0),
    Voter(id='V4', x=0.0, r=0.0)
```

### Experiment 2 - large profiles

For the larger size of profiles we generated a suitable number of random VCR profiles and checked how many of them are not VR nor CR. To create random VCR profile we generated points $x_a$ and radii $r_a$ for every agent $a \in C \cup V$ and then created preference matrix based on the VCR definition.

Additionally to have a broder view of the domain the experiment is split into sub-experiments based on the different random distributions (WIP: Czy "distributions" jest ok?) used to generate profiles.

### Uniform distribution

For this sub-experiment we generated points and radii with uniform distribution from the ranges $[0, 750)$ and $[0, 100)$ respectively.

(WIP: Napisać jaki wpływ mają parametry na "strukturę" profilu -> np dla 20x20. $sum(sum(A)) \in [180, 250]$)

| | 10C 10V | 15C 15V | 20C 20V |
|---|---|---|---|
| VCR | 1000 | 1000 | 3000 |
| VR | - | - | - |
| CR | - | - | - |
| VCR !COP | 24 | 667 | 2928 |

Table 5.2 Temporary table - Large profiles NCOP/VCR

### Some other distribution

WIP:

# Chapter 6

# HD-Minimax Scoring Rule Algorithms

## 6.1 Hamming Distance Minimax Approval Voting

### Hamming Distance

Given two sets $X, Y$ their Hamming distance is $d_{Ham}(X,Y) = |X \setminus Y| + |Y \setminus X|$. Alternatively for a vector representation we define the Hamming distance between vectors of the same size $a, b$ as $d_{Ham}(a,b) = |a \oplus b|$.

### Minimax Approval Voting (MAV)

Scoring rule introduced by Brams et al. (2007). Selects a committees $W$ that minimizes the maximum Hamming distance between $W$ and voters' approval ballots, i.e. *MAV mininimizes* $max_{v \in V}\ d_{Ham}(A(v), W)$. Intuitively MAV follows the egalitarian principle (WIP: Co to dokładnie / czy istnieje definicja?), it selects committees that minimize the dissatisfaction of the least happy voter.

(WIP: poprawić zdanie)

### (WIP) Computational complexity

Computing the winning committee for MAV is known to be NP-hard, see **?** (WIP: cytacje). Nevertheless, Liu and Guo (2016) provide polynomial-time algorithms for computing MAV results, for the special case of the elections with candidate range or voter range properties.

(WIP: Poprawić zdanie)

## 6.2 (WIP) My algorithm for VCR profiles

**( WIP ) Current approches**

**Greedy**

1. Sort by ?
2. ?

**Dynamic**

1. Recursive
2. branch and bound
3. ?

**Augumented Data** Structure

1. ?

# Chapter 7

# (WIP) Approval deletion for VCR profile to become VR or CR

# Chapter 8

# Control Problem

## 8.1 Overview

## 8.2 Control for the VCR profiles

# Chapter 9

# Evaluation

## 9.1 Conclusions

## 9.2 Future Work

# References


Brams, S., Kilgour, D., and Sanver, M. R. (2007). *A Minimax Procedure for Negotiating Multilateral Treaties*, pages 265–282.

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Liu, H. and Guo, J. (2016). *Parameterized Complexity of Winner Determination in Minimax Committee Elections*, page 341–349. AAMAS '16. International Foundation for Autonomous Agents and Multiagent Systems.